

Unknown Disturbance Compensation Control of PMSM based on Extended State Observer

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Abstract—There are many kinds of disturbances in the actual permanent magnet synchronous motor (PMSM) control system, including load sudden change, friction and others, which affect its speed control performance. The unknown torque disturbance and load torque in the PMSM system are considered to total disturbance using disturbance compensation control strategy in this paper, the extended state observer (ESO) is used to estimate the total disturbance in real time, and the feedforward compensation is applied to the speed loop controller to suppress the system disturbance. The trapezoidal wave disturbance and square wave disturbance are introduced in the PMSM system for simulation of the disturbance estimation and speed response, the estimation effects of total disturbance through output torque and input q-axis current are compared. The results show that the disturbance compensation control based on ESO can effectively suppress the unknown system disturbance, improve the speed tracking accuracy and enhance the robustness of the system.

Index Terms—Permanent magnet synchronous motor(PMSM), Disturbance estimation, Extended state observer(ESO), Compensation control

I. INTRODUCTION

PMSM is widely used in industry, transportation, military and aviation because of its simple structure and high efficiency[1]. However, PMSM has the characteristics of strong coupling, variable parameters and non-linearity. In order to obtain a PMSM control system with high stability, good dynamic performance and high control accuracy, the excellent anti-disturbance performance becomes the key.

Different control strategies were put forward to improve the speed control performance of PMSM control system. The second-order non-linear active disturbance rejection control (ADRC) was designed to observe the load disturbance in position-speed loop compound control [2], results improved the anti disturbance performance. The current loop adopted the first-order linear ADRC to replace the traditional PI method

to improve the dynamic response speed, this method achieved better control effect. [3] combined sliding mode variable structure with active disturbance rejection control strategy, designed the sliding film feedback control rate to load on the speed loop, improved the torque characteristics and reduced the speed overshoot. [4] designed the motor model into known part and unknown part, and compensated for the known quantity into ADRC, which improved the speed estimation accuracy and the speed performance. Fuzzy control was combined with ADRC to modify parameters online and in real time [5], though it achieved certain results, the control algorithm and the calculation were complex. [6] applied single neuron PID to the speed loop, combined with the idea of feedforward compensation and fuzzy control, applied feedforward compensation to the load torque, which improved the anti-disturbance ability and stability of the system. To compensate the system disturbance in real time, a new speed tracking control method was proposed combining generalized predictive control theory and ESO in reference [11]. In reference [12], a controller composed of nonlinear ESO, tracking differentiator and dynamic surface control was proposed to suppress the disturbance, it prevented the chaotic parameters of PMSM from falling into some special regions. Different from the existing PMSM control methods, this method has high transient and steady-state stability. The nonlinear ESO was used to estimate the unmeasurable state and uncertainty of the system, the tracking performance was significantly improved combined with the tracking differentiator, and the anti-interference ability of the output of the PMSM was improved in the presence of uncertainty and unmodeled dynamics. In reference [13], an advanced predictive control method based on Lyapunov theory was developed, which used the multivariate approximation of quadratic constraints to reduce the complexity. The piecewise

control law is obtained by multi parameter linear programming, and the state and disturbance observers were designed by using quadratic Lyapunov function to ensure the asymptotic stability of the estimation error system. In reference [14], a new robust control scheme with three first order was proposed for PMSM speed control. Compared with the existing high-order ADRC speed control structure, this method did not need to estimate the rotor flux.

Based on vector control and PID control, ESO is introduced into speed loop to estimate and compensate for the unknown disturbance in real time, so as to improve the dynamic performance of PMSM control system. We introduced the square wave disturbance and trapezoidal wave disturbance to verify the estimation and compensation ability of the proposed method, and compared the disturbance estimation values obtained by torque and current are compared.

II. METHOD

A. Mathematical Model of PMSM

If the eddy current loss and hysteresis loss of PMSM are neglected, assume the magnetic circuit to be unsaturated, the voltage equation of PMSM in d-q rotating coordinate system is as follows:

$$\begin{cases} u_d = \frac{d\psi_d}{dt} - \omega_e \psi_q + R_s i_d \\ u_q = \frac{d\psi_q}{dt} + \omega_e \psi_d + R_s i_q \end{cases} \quad (1)$$

where u_d, u_q are the stator voltage of d axis and q axis; i_d, i_q are stator current of d axis and q axis; ψ_d, ψ_q are the flux of the d axis and q axis; R_s is the stator resistance; ω_e is the angular velocity of the rotor.

Stator flux linkage equation,

$$\begin{cases} \psi_d = L_d i_d + \psi_f \\ \psi_q = L_q i_q \end{cases} \quad (2)$$

where L_d, L_q are the inductance of d-axis and q-axis respectively, ψ_f is the rotor flux.

Electromagnetic torque equation,

$$T_e = \frac{3}{2} n_p \vec{\psi}_s \times \vec{i}_s = \frac{3}{2} n_p [\psi_f i_q + (L_d - L_q) i_d i_q] \quad (3)$$

where n_p is the number of poles of the motor. The second term of the formula is the reluctance torque, but for the surface mounted permanent magnet synchronous motor (SPMSM), the direct axis inductance L_d and the quadrature axis inductance L_q are equal, so the torque formula of the SPMSM is as follows:

$$T_e = \frac{3}{2} n_p \vec{\psi}_s \times \vec{i}_s = \frac{3}{2} n_p \psi_f i_q \quad (4)$$

The equation of motion is as follows:

$$T_e - T_L = J \frac{d\omega}{dt} + B\omega \quad (5)$$

$$\frac{d\omega}{dt} = \frac{1}{J} \left[\frac{3}{2} n_p \psi_f i_q - T_L - B\omega \right] \quad (6)$$

where T_e is the electromagnetic torque, T_f is the load torque, ω is the mechanical angular velocity of the motor, B is the viscous friction coefficient, J is the load moment of inertia.

B. Estimation and compensation for unknown disturbance

In order to improve the dynamic performance of the system, it is necessary to compensate for the disturbance caused by friction and modeling error. In this paper, an extended state observer, which is not based on the accurate system model, is proposed to compensate for the disturbance in the system. The speed loop control model of the system is shown in Figure 1.

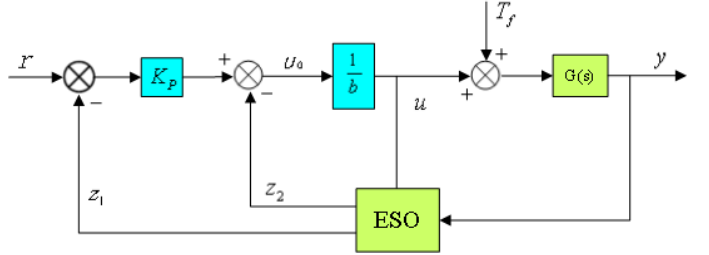


Fig. 1. System speed loop control scheme.

In Fig. 1, K_P, b are the controller parameters of the speed loop respectively; r and y are the input and output of the speed loop; T_f is the disturbance in the system; u_0 is the input signal of the control object before compensation; u is the input signal of the control object after compensation; $G(s)$ is the control object of the speed loop; ESO is an extended state observer; z_1 and z_2 are the estimated results of ESO.

Because there will always be unknown disturbance torque in the system, according to the previous analysis, the equation of motion can be transformed into the form of (7),

$$\dot{\omega}_m = \frac{T_e}{J} - \frac{T_l}{J} - \frac{T_f}{J} \quad (7)$$

where T_l represents load torques of the system.

Rewriting the equation of motion, the unknown disturbance and load torque are attributed to unified external disturbance and internal disturbance, and the following equation can be obtained,

$$\begin{aligned} \dot{x} &= bu + f(t, \omega_m) \\ x &= \omega_m \\ f(t, \omega_m) &= \frac{T_e}{J} - bu - \frac{T_f}{J} - \frac{T_l}{J} \end{aligned} \quad (8)$$

$f(t, \omega_m)$ represents the total disturbance, $T_e/J - bu - T_f/J$ is the internal disturbance, $-T_l/J$ is the external disturbance, b is an adjustable parameter in the speed loop.

Here we assume $f(t, \omega_m)$ is differentiable and that there are $h(t) = df(t, \omega_m)/dt, |h(t)| \leq \beta_m$. Moreover, there is a positive number $f(\beta_m)$ that satisfies this relationship. Then the total disturbance in the system $f(t, \omega_m)$ can be expanded to a separate state. The system will be converted to the following expression,

$$\begin{aligned} \dot{x}_1 &= bu + x_2 \\ \dot{x}_2 &= -h \end{aligned} \quad (9)$$

where $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T = \begin{bmatrix} \omega_m & f \end{bmatrix}^T$.

According to the equation (9), there will be the following state-space equations,

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \quad (10)$$

where $\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}$, $y = x_1 = \omega_m$, $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} b & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

Therefore, the observation matrix of the system will have such a relationship:

$$\text{rank} \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 2 \quad (11)$$

Equation (11) indicates that the system is observable. Therefore, such an ESO for a system can be constructed:

$$\begin{aligned} \dot{\hat{x}}_1 &= bu + \hat{x}_2 - \beta_0 (\hat{x}_1 - x_1) \\ \dot{\hat{x}}_2 &= -\beta_1 (\hat{x}_1 - x_1) \end{aligned} \quad (12)$$

β_0, β_1 are the design parameter of the extended state observer in the inner speed loop.

Thus, $\eta_i = \hat{x}_i - x_i, i = 1, 2$, which represents the observation error of the extended state observer, and equation (9) minus equation (12) yields the following equation:

$$\dot{\eta} = A\eta + Mh \quad (13)$$

where $\eta = \begin{bmatrix} \eta_0 \\ \eta_1 \end{bmatrix}$, $A = \begin{bmatrix} -\beta_0 & 1 \\ -\beta_1 & 0 \end{bmatrix}$, $M = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The above formula shows that if the matrix A is Hurwitz, the extended state observer is BIBO stable. In a matrix A , if its design parameters β_0, β_1 are designed to be:

$$\beta_0 = 2\omega_{m0}, \beta_1 = \omega_{m0}^2 \quad (14)$$

Here, ω_{m0} is the bandwidth of the extended state observer in the inner loop, which is a parameter to be designed. Equation (14) guarantees the BIBO stability of the system.

If $\eta_i(t)$ is bounded, then there will be a positive number and a finited time to make,

$$|\eta_i^m(t)| \leq \zeta_i, \zeta_i = o\left(\frac{1}{\omega_{m0}^c}\right); i = 1, 2 \forall t \geq T_1 \quad (15)$$

Equation (15) makes the constructed extended state observer a BIBO stability observer. The results of the lemma show that the extended state observer has an excellent estimation ability for disturbances. Its estimation ability depends on its bandwidth ω_{m0} . When ω_{m0} is large enough, it can accurately estimate the states $x_1^m(t)$ and $f(t, \omega_m)$ sum of the system.

From the analysis of the lemma, it is known that when an appropriate parameter ω_{m0} is designed, it can make the $o\left(\frac{1}{\omega_{m0}^c}\right) \approx 0$. There will be $|x_2 - \hat{x}_2| = |\eta_2(t)| \leq o\left(\frac{1}{\omega_{m0}^c}\right)$, that is $\hat{x}_2 \approx f(t, \omega_m)$.

Therefore, let $u = (u^0 - \hat{x}_2)/b$ and then replace it with equation (8),

$$\left. \begin{aligned} \dot{x} &= b[(u^0 - \hat{x}_2)/b] + f(t, \omega_m); \\ x &= \omega_m. \end{aligned} \right\} \Rightarrow \begin{aligned} \dot{x} &= u^0 - [\hat{x}_2 - f(t, \omega_m)] \\ x &= \omega_m \end{aligned} \quad (16)$$

Above formula indicates that the nonlinear system (8) will be transformed into an approximate linear integral system (17).

$$\begin{aligned} \dot{x} &\approx u^0 \\ x &= \omega_m. \end{aligned} \quad (17)$$

By reasonable parameter designing, the disturbance in the system can be eliminated.

III. RESULTS

We analyze the algorithm of PMSM to verify the feasibility, and the parameters of PMSM are selected as follows:

TABLE I
MOTOR PARAMETERS

Resistance	0.9(ohm)
Inductance	0.000835(H)
Flux	0.23125(V.s)
Coefficient of torque	2.775(N.m/A_peak)
Moment of inertia	0.014(kg.m ²)
Damping coefficient	0.00425(N.m.s)
Number of pole pairs	8

Given a speed signal as 100r/min, a trapezoidal disturbance with an amplitude of $T_l = 20N.m$ is applied to the system at $t=0.015s$ and a square disturbance with $T_l = 30N.m$ is applied at $t=0.025s$. The system disturbance estimation compensation capability is tested for trapezoidal and square disturbances, respectively.

A. Analysis of Trapezoid Wave Disturbance Compensation Performance

The green line in Figure 2 represents the given speed, the black line represents the tracking curve of the actual speed without compensation, and the red line represents the tracking curve of the actual speed after the disturbance is compensated. After the actual speed reaches 100r/min and reaches steady state, 20N.m trapezoidal disturbance is applied at 0.015s. When there is no compensation for the disturbance, the speed drops to 97 r/min, and after the disturbance disappears, the system tracks the given speed again. When ESO is used to compensate for the disturbance, the speed fluctuation is within 1 r/min. From Figure 2, it can be seen that the disturbance observer can accurately estimate the applied disturbance, so as to effectively compensate for the disturbance and improve the system speed control performance.

In Figure 3, the red line represents the applied trapezoidal perturbation torque, and the blue line represents the torque estimated by ESO. The figure3(a) estimates the disturbance torques by the motor output torques, while the figure3(b) estimates the disturbance torques by the input q-axis current. The maximum error between the estimated value and the actual value is not more than 2.5 N.m when the output torque

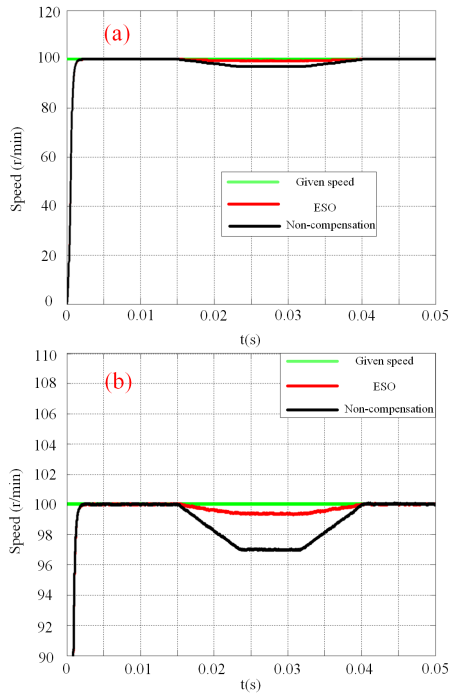


Fig. 2. Comparison of 100r/min speed tracking results under trapezoidal wave disturbance .

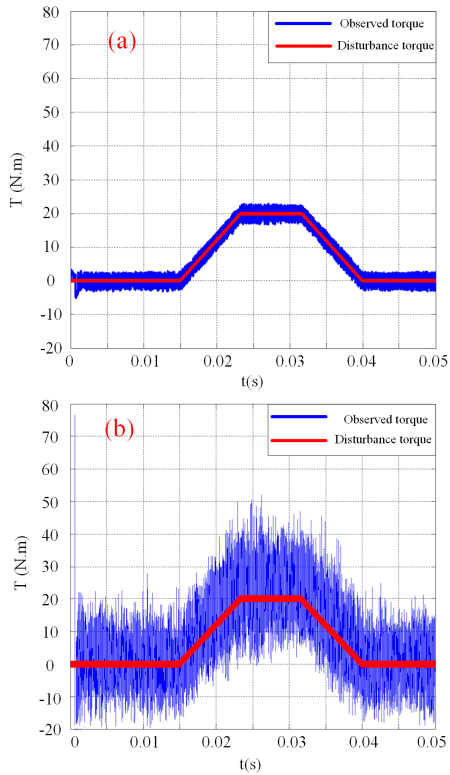


Fig. 3. Estimation results of trapezoid wave disturbance for current and torque.

is used for disturbance observation. When the q-axis current is used for disturbance observation, the maximum estimation error is 20N.m. The graph shows that although the q-axis current is proportional to the output torque, some non-linear noise is introduced. If there is a torque sensor, the load torque can be estimated by the torque sensor, which has a higher degree of restore to the disturbance. However, it is also possible to estimate the disturbance torque by q-axis current.

B. Analysis of Square Wave Disturbance Compensation Performance

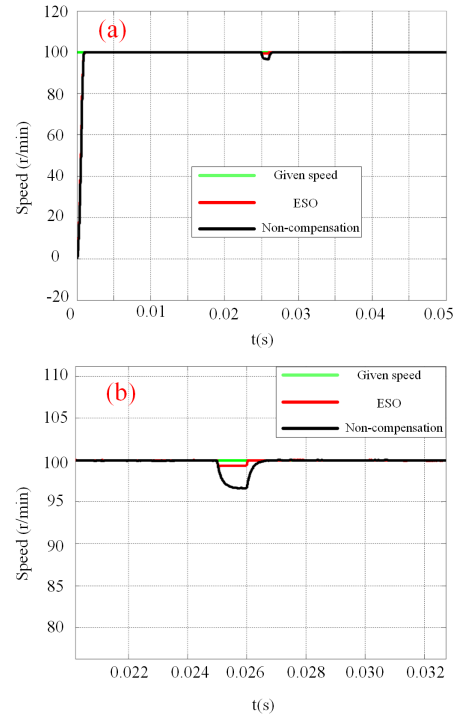


Fig. 4. Comparison of 100r/min speed tracking results under square wave disturbance.

Figure 4 is similar to the result of Figure 2, in which a 30N.m square wave disturbance is applied to the motor when it runs smoothly at 100r/min. When there is no compensation for the disturbance, the speed fluctuation is 3r/min, 0.5ms after the disturbance disappears, the system tracks the given speed again. When ESO is used to compensate the disturbance, under the same circumstances, the speed fluctuation is within 1 r/min, and the given speed is tracked immediately after the disturbance disappears, with the adjustment time being almost zero. Similar to the results from Fig. 5 and Fig. 3, the maximum estimation error is approximately 2.5 N.m when the output torque is used for the disturbance observation. When the q-axis current is used for the perturbation observation, while the estimation error is 25N.m. Moreover, the estimation error is greater at the rising and falling edge of square wave. Figures 4 and 5 show that the disturbance observer used in this paper can also effectively estimate and compensate disturbance torques under square wave disturbance. Similarly,

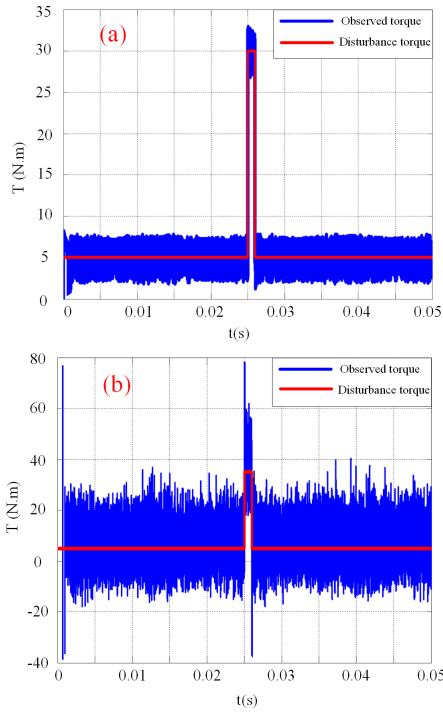


Fig. 5. Estimation results of square wave disturbance for current and torque.

the disturbance torques observed by output torques are more restorative than those observed by q-axis current.

In summary, results show that the disturbance observer can estimate and compensate for different types for disturbances. In practical control systems, we usually do not know the type of disturbance, we can effectively compensate for the unknown disturbance in the system by the proposed method.

The specific values of speed tracking results under trapezoidal wave and square wave disturbances are shown in Table 2. When there is no disturbance in the system, the effect of disturbance observer compensation is not reflected, even the mean square error is high. This is because the disturbance observer introduced some noise, but the effect is not significant. By comparing the estimated compensation results of trapezoidal and square wave disturbances, it can be seen that under trapezoidal wave disturbance, the speed mean square error decreases from 1.2563 to 0.2633; Under the square wave disturbance, the mean square error of speed decreases from 0.4786 to 0.0998.

TABLE II
SPEED ERROR COMPARISON

	(r/min)	Maximum error	Mean square error
No disturbance	Non-compensation	0.0744	0.0189
	Observer	0.0744	0.0198
Trapezoidal disturbance	Non-compensation	3.0370	1.2563
	Observer	0.6585	0.2633
Square wave disturbance	Non-compensation	3.3574	0.4786
	Observer	0.7410	0.0998

IV. CONCLUSION AND DISCUSSION

Based on the PMSM vector control system, an extended state observer is introduced into the speed loop controller. The estimation ability and noise suppression performance of ESO for different types of disturbances are analyzed, and the difference between the disturbance estimates obtained by q-axis current and output torque is compared. The following conclusions are drawn:

(1) For trapezoidal and square wave disturbances, ESO-based disturbance compensation controller has excellent anti-disturbance ability, the speed control accuracy has been improved nearly five times. When the system is steady, the observer itself produces weak noise, however, it has little impact with estimation error.

(2) Compared with the disturbance estimation value obtained from q-axis current, the accuracy of the disturbance estimation value obtained from output torque is 8-10 times that of q-axis current estimation value. Using output torque as the input of observer can better reflect the real disturbance of PMSM control system.

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